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# The Energy of Memorization: Why Graph-Level Label Noise Sharpens Representations and How to Stop It

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## Abstract

1 Graph Neural Networks (GNNs) are vulnerable to label noise in graph classification,  
2 yet the mechanisms driving this vulnerability are poorly understood. We empirically  
3 identify a property of the *learned representations* that tracks noise memorization:  
4 the within-graph Dirichlet energy of node features. Energy decreases as the model  
5 captures genuine class structure, then rises sharply, mainly in high-frequency  
6 components, when the model begins to memorize corrupted graph labels. We  
7 support this observation with a Poincaré-based lower bound on the energy implied  
8 by noise fitting, combined with an entropy-based asymmetry result adapted from  
9 Qian et al. [2025] indicating that the energy growth is preferentially absorbed  
10 by graphs most informative for classification. Guided by these observations we  
11 design and evaluate three interventions: a Dirichlet-energy regularizer ( $\mathcal{L}_{DE}$ ), a  
12 robust loss (GCOD), and a spectral weight constraint (CE+W2). Across eight  
13 benchmarks, including the full and modified ogbg-ppa, symmetric noise up to 60%,  
14 asymmetric noise, and a transfer to Cora node classification, the three interventions  
15 all reduce within-graph energy and improve robustness while preserving clean-  
16 data performance. GCOD, our practical recommendation, matches or exceeds  
17 OMG [Yin et al., 2023] at lower training overhead.

## 18 1 Introduction

19 Graph Neural Networks (GNNs) are now the workhorse of *graph classification*, the task of assigning  
20 a single label to an entire graph, with applications spanning molecular property prediction [Wale and  
21 Karypis, 2006], bioinformatics [Borgwardt et al., 2005], and social-network analysis [Yanardag and  
22 Vishwanathan, 2015a]. In real deployments, the labels attached to graphs are routinely noisy: assays  
23 disagree, annotators err, and weak supervision injects systematic mistakes. Although learning under  
24 label noise has been thoroughly studied for images [Algan and Ulusoy, 2021] and for node classifica-  
25 tion [Dai et al., 2021, Yuan et al., 2023a], graph classification under noisy labels is dramatically less  
26 explored [NT et al., 2019, Yin et al., 2023].

27 **Graph classification is a fundamentally different setting.** In node classification, label noise  
28 corrupts *individual node labels* on a single graph, and homophily, the tendency of connected nodes  
29 to share labels, is a structural prior that directly couples Dirichlet energy of the *label signal* to  
30 noise. In graph classification, label noise flips the label of an *entire graph*: nodes have no individual  
31 classification labels, and within-graph homophily/heterophily is largely orthogonal to the graph-level  
32 label. The standard tools and intuitions from node classification therefore do not transfer.

**Our central observation.** We measure the Dirichlet energy of *learned node representations*, not of labels. Empirically, this energy first decreases as the model captures genuine, transferable patterns; then, when the model begins to memorize incorrect graph-level labels, it rises sharply, with the surge concentrated in the high-frequency spectrum. This rise is not a tautology: noise is injected only at the graph-label level, while node features and edges are unchanged, so any correlation between energy and labels has to emerge through training dynamics. Theorem 1 supports this observation analytically: under a lower-Lipschitz regularity assumption on the encoder, fitting noisy graph-level labels for two structurally similar graphs forces a lower bound on their within-graph Dirichlet energies via the Graph Poincaré inequality.

**Why does suppressing this energy growth help?** Combining a dataset-level spectral identity (Proposition 1) with an entropy argument from Qian et al. [2025], Section 6 argues that uncontrolled energy growth is preferentially absorbed by *high-entropy graphs*, the ones carrying most discriminative signal at pooling. Reducing within-graph Dirichlet energy therefore plausibly suppresses noise fitting while preserving the high-entropy regions classification accuracy depends on, a hypothesis we test with three mechanistically distinct interventions.

## Contributions.

- A controlled study of *when* GNNs fit graph-level label noise (Section 4), isolating three failure modes: over-parameterization, low-order graphs, and small training sets.
- Within-graph Dirichlet energy as a diagnostic for graph-level noise memorization (Section 5), with a frequency decomposition that localizes the surge to high-frequency components and an empirical scaling with the noise rate.
- Analytical support for the diagnostic (Section 6): a spectral identity (Proposition 1), a Poincaré-based lower bound under noise fitting (Theorem 1), and an entropy-based asymmetry result (Corollary 1). These results are unique to graph classification, where labels and feature smoothness live on different objects, unlike node classification.
- Three interventions (Section 7): a Dirichlet-energy regularizer  $\mathcal{L}_{DE}$ , a centroid-based loss GCOD, and a spectral weight constraint CE+W2. They are mechanistically distinct and all reduce within-graph energy while improving robustness; GCOD is our practical recommendation.
- Evaluation on eight benchmarks under symmetric, asymmetric, and high-rate (60%) noise, the full and a modified ogbg-ppa, and a transfer to Cora node classification. GCOD matches or exceeds OMG [Yin et al., 2023] and SOP [Liu et al., 2022] at  $\sim 70\times$  lower compute.

## 2 Related Work

**Learning under label noise.** Robust loss functions [Ghosh et al., 2017, Patrini et al., 2017], sample re-weighting [Liu and Tao, 2015, Liu et al., 2022], early-learning regularizers [Arpit et al., 2017], and small-loss filtering [Gui et al., 2021] dominate the image-classification literature. Wani et al. [2023] introduce NCOD, a centroid-based outlier-discounting loss that we adapt to graphs.

**Noisy-label learning on graphs.** Most prior work concerns *node* classification, exploiting homophily, contrastive losses, or pseudo-labels [Dai et al., 2021, Yuan et al., 2023a,b, Li et al., 2024, Kang et al., 2018]. For *graph* classification, NT et al. [2019] propose a surrogate loss that discards estimated-noisy labels under specific assumptions, and OMG [Yin et al., 2023] combines contrastive learning, MixUp [Lim et al., 2022], and curriculum filtering. We differ in two ways: (i) we provide a spectral/energy-based explanation of *why* GNNs fit graph-level noise, rather than only mitigating it, and (ii) our practical method (GCOD) achieves better or comparable robustness at a fraction of the compute.

**Dirichlet energy and over-smoothing.** Dirichlet energy quantifies signal smoothness on graphs [Zhou and Schölkopf, 2005]. Existing analyses establish that GNNs act as low-pass filters whose energy decays exponentially with depth under spectral conditions [NT and Maehara, 2019, Cai and Wang, 2020, Oono and Suzuki, 2020, Rusch et al., 2023], and that weight-matrix eigenvalues control attraction/repulsion of node features [Giovanni et al., 2023]. Energy-based regularizers [Zhou et al., 2021, Bo et al., 2021, Chen et al., 2023] address *too little* energy (over-smoothing) on *node* classification. We address the opposite regime: *too much* within-graph energy caused by graph-level label noise. Qian et al. [2025] recently show that, for graph classification specifically, over-smoothing in *high-entropy* regions is the most damaging; we leverage this insight (Corollary 1) to argue why

suppressing energy growth is essential under graph-level noise. A more comprehensive related-work discussion is in Appendix A.

### 3 Background

**Notation.** A graph is  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$  with  $N = |\mathcal{V}|$ , degree matrix  $\mathbf{D}$ , adjacency  $\mathbf{A} \in \{0, 1\}^{N \times N}$ , feature matrix  $\mathbf{X} \in \mathbb{R}^{N \times m}$ , combinatorial Laplacian  $\mathbf{L} = \mathbf{D} - \mathbf{A}$  (with eigenvalues  $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ ) and normalized Laplacian  $\Delta = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$ . We write  $\lambda_2(\mathcal{G})$  for  $\lambda_2(\mathbf{L})$  throughout and use  $\Delta$  only when explicitly stated. In *graph classification*, the dataset is  $\mathcal{D} = \{(\mathcal{G}^i, \mathbf{y}_i)\}_{i=1}^n$ ,  $\mathbf{y}_i \in \{0, 1\}^{|C|}$  the one-hot graph label; we write  $c_i$  for the class. Under message passing,  $\mathbf{H}_i^{l+1} = \text{UP}_{\Omega_i}(\mathbf{H}_i^l, \text{AGGR}_{\mathbf{W}_i}(\mathbf{H}_i^l, \mathbf{A}^i))$ , with  $\mathbf{H}_i^0 \equiv \mathbf{X}^i$  and  $\mathbf{Z}^i \equiv \mathbf{H}_i^L$ . A permutation-invariant readout  $f_\theta : \mathbb{R}^{N^i \times m'} \rightarrow \mathbb{R}^{|C|}$  maps node features to class probabilities  $\hat{\mathbf{y}}_i$ . Under noisy labels, the observed  $c_i$  may differ from the ground truth; the test set is clean.

**Dirichlet energy on graphs.** For graph  $\mathcal{G}^i$  with representation  $\mathbf{Z}^i \in \mathbb{R}^{N^i \times m'}$  we use the combinatorial Dirichlet energy

$$E^{\text{dir}}(\mathbf{Z}^i) = \sum_{(u,v) \in \mathcal{E}^i} \|\mathbf{Z}_u^i - \mathbf{Z}_v^i\|_2^2 = \text{tr}((\mathbf{Z}^i)^\top \mathbf{L}^i \mathbf{Z}^i). \quad (1)$$

Small  $E^{\text{dir}}$  means smooth (low-frequency) representations within the graph; large  $E^{\text{dir}}$  means sharpening (high-frequency). The symmetrically-normalized variant  $E_{\text{sym}}^{\text{dir}}(\mathbf{Z}) = \sum_{(u,v)} \|\mathbf{Z}_u / \sqrt{d_u} - \mathbf{Z}_v / \sqrt{d_v}\|_2^2 = \text{tr}(\mathbf{Z}^\top \Delta \mathbf{Z})$ , used in some over-smoothing analyses [NT and Maehara, 2019, Oono and Suzuki, 2020], is equivalent up to degree-bounded constants and our qualitative claims hold for either; we use (1) because it pairs cleanly with the Poincaré inequality of Section 6. To track behavior across a class-restricted subset  $\mathcal{S}$  of training graphs we use

$$E_{\text{set}}^{\text{dir}}(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{\mathcal{G}^i \in \mathcal{S}} E^{\text{dir}}(\mathbf{Z}^i). \quad (2)$$

Crucially, (1) is computed on *learned representations*, not on labels. Within-graph homophily of features therefore evolves with training, decoupled from the graph-level label of  $\mathcal{G}^i$ .

### 4 When GNNs Fit Graph-Label Noise

GNNs trained with standard cross-entropy (CE) on graph classification tasks are not uniformly vulnerable to label noise. On the full ogbg-ppa [Szklarczyk et al., 2018a] (37 classes, the standard OGB benchmark), even 40% symmetric noise barely degrades CE accuracy, because the model is under-parameterized for the full task and behaves robustly [Zhang et al., 2021]. To expose the noise-fitting regime, we control three axes: **(F1)** number of classes (proxy for over-parameterization), **(F2)** graph order (size of each graph), and **(F3)** training-set size. We use a 5-layer GIN [Xu et al., 2019] with 300 hidden units throughout, unless stated.

In all three regimes (Fig. 1 and Appendix D), GNNs progressively memorize noisy graph-level labels. The natural next question is therefore: what changes in the network’s internal representations during memorization, and can this change be detected and controlled? The next section identifies Dirichlet energy as that internal signal.

### 5 Dirichlet Energy as a Diagnostic for Noise Fitting

We track  $E_{\text{set}}^{\text{dir}}(\mathcal{D}_c)$ , the within-graph energy averaged over training graphs of class  $c$ . Across architectures and datasets (Fig. 2), the diagnostic is consistent:  $E_{\text{set}}^{\text{dir}}$  falls during the early “learning” phase, then rises precisely when the model begins to fit noisy graphs (CE-20% in Fig. 2b), with 40% noise yielding a strictly larger end-of-training energy than 10% noise (Appendix F).

**Energy is sensitive to noise specifically through high-frequency components.** A natural concern is that energy may also rise during clean-data overfitting [Yan et al., 2022]; we confirm this in Appendix F but show the noise-driven surge is markedly steeper. To localize the surge spectrally, we

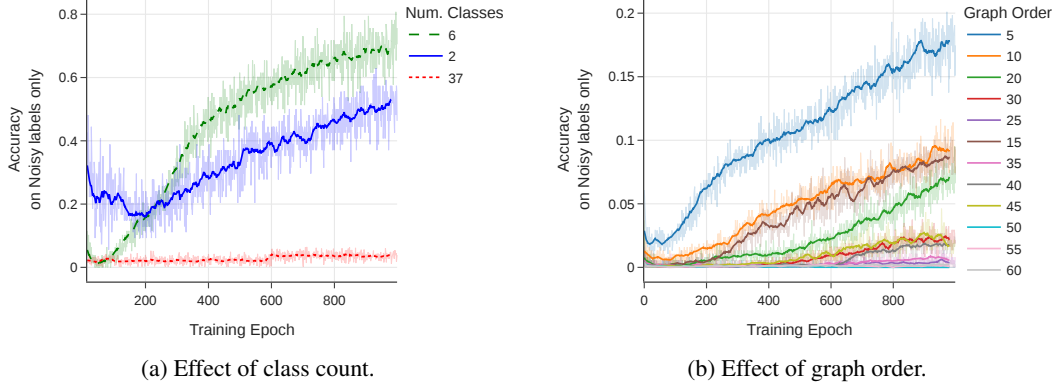


Figure 1: Training accuracy on *noisy labels only*: a higher curve means more memorization. (a) Reducing the number of PPA classes (with 20% symmetric noise) makes a fixed GIN over-parameterized, accelerating noise fitting. (b) Smaller average graph order (synthetic data, 35% noise) leaves less internal structure for averaging, enabling memorization.

127 use the HLFF-GNN decomposition [Xu et al., 2024] to split representations into low-frequency ( $Z_1$ )  
 128 and high-frequency ( $Z_2$ ) components. On ENZYMES with 30% noise,  $E^{\text{dir}}(Z_1)$  remains essentially  
 129 unchanged, while  $E^{\text{dir}}(Z_2)$  rises monotonically, and slope/variance statistics of  $E^{\text{dir}}(Z_2)$  provide an  
 130 early warning for label noise (Appendix E). HLFF-GNN is used purely as an analytical tool; the same  
 131 total-energy surge appears in standard GIN/GCN.

132 **Why energy is more useful than a generic overfitting signal.** Standard noise-detection signals  
 133 require a *clean* held-out set: validation loss is uninformative when validation labels follow the same  
 134 noise distribution as training (the noisy validation loss continues to decrease as the model fits noise,  
 135 as we observe in our setting); train–test gap is unobservable at training time; small-loss heuristics  
 136 need a calibration regime. Within-graph Dirichlet energy is computed entirely on the training graphs  
 137 and their representations, so it remains a valid noise-fitting indicator under fully-noisy supervision.  
 138 Beyond replacing the missing clean signal, it tells you *how* the network is failing: which spectral  
 139 components drive the surge. This mechanistic specificity is what turns a diagnostic into a design  
 140 principle, and motivates the three interventions in Section 7, each of which targets the same harmful  
 141 spectral behavior in a distinct way.

## 142 6 Analytical Support: Why Energy Suppression Helps

143 Section 5 established Dirichlet energy as a reliable empirical signal. We now show this is not just  
 144 correlation: under standard regularity, within-graph energy is *provably forced* to grow when a GNN  
 145 fits graph-level label noise, with the growth absorbed disproportionately by structurally rich graphs.  
 146 The argument combines three pieces: a dataset-level spectral identity (Proposition 1), the Graph  
 147 Poincaré inequality (Lemma 1), and a lower-Lipschitz bound on the GNN (Lemma 2) [Chuang and  
 148 Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024], yielding Theorem 1.

### 149 6.1 Three Building Blocks

150 **Proposition 1** (Dataset-level Dirichlet energy). *Let  $\mathcal{D} = \{\mathcal{G}^1, \dots, \mathcal{G}^n\}$  with representations*  
 151  *$\mathbf{Z}^i \in \mathbb{R}^{N^i \times m'}$  and adjacencies  $\mathbf{A}^i$ . Define the block-diagonal graph  $\bar{\mathcal{G}} = (\bar{\mathbf{Z}}, \bar{\mathbf{A}})$  with  $\bar{\mathbf{Z}} =$*   
 152  *$[\mathbf{Z}^1 \parallel \dots \parallel \mathbf{Z}^n] \in \mathbb{R}^{N_{\text{tot}} \times m'}$  (vertical concatenation) and  $\bar{\mathbf{A}} = \text{blkdiag}(\mathbf{A}^1, \dots, \mathbf{A}^n)$ , with associated*  
 153 *combinatorial Laplacian  $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$ . Then*

$$E_{\text{set}}^{\text{dir}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}}) = \frac{1}{|\mathcal{D}|} \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2, \quad (3)$$

154 *where  $\{(\bar{\lambda}_u, \bar{\psi}_u)\}$  are eigenpairs of  $\bar{\mathbf{L}}$ , and the spectrum decomposes as  $\bar{\mathbf{L}} = \bigcup_i \Lambda^i$  with each*  
 155 *eigenvector supported on a single connected component  $\mathcal{G}^i$  (graph or block).*

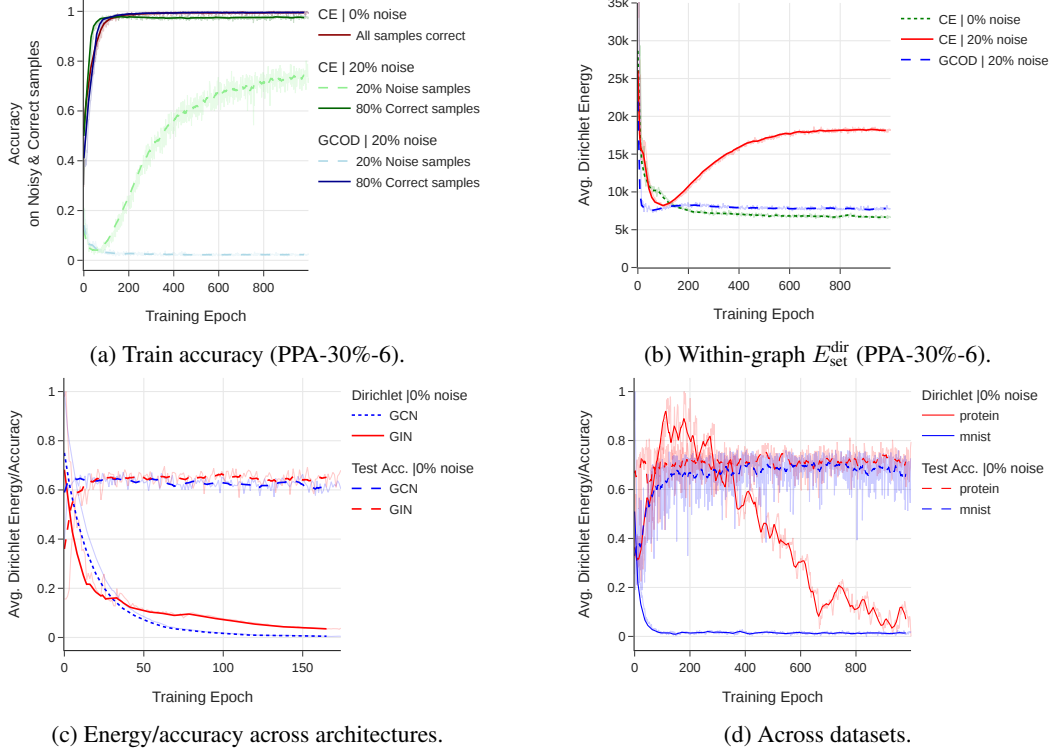


Figure 2: Within-graph Dirichlet energy of learned representations *rises during noise memorization*, across architectures (GIN/GCN) and datasets (PPA, MUTAG, MNIST, PROTEINS). Under clean labels,  $E_{\text{set}}^{\text{dir}}$  steadily decays as accuracy improves; under noise, it follows the same decay until the model starts fitting noisy graphs, after which it sharply rises. GCOD suppresses this surge.

The proof (Appendix B) follows from the block-diagonal structure of  $\bar{\mathbf{L}}$ . The key structural fact: *each high-frequency eigenvector of  $\bar{\mathbf{L}}$  is owned by exactly one graph*, so sharpening along it raises only that graph’s energy.

**Lemma 1** (Graph Poincaré, vector-valued form). *For any feature matrix  $\mathbf{Z} \in \mathbb{R}^{N \times m'}$  on a connected graph  $\mathcal{G}$  with combinatorial Laplacian  $\mathbf{L} = \mathbf{D} - \mathbf{A}$  and second-smallest eigenvalue  $\lambda_2(\mathcal{G}) > 0$ , let  $\bar{\mathbf{z}} = \frac{1}{N} \sum_v \mathbf{Z}_v$ . Then*

$$\sum_{v=1}^N \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} E^{\text{dir}}(\mathbf{Z}). \quad (4)$$

(4) bounds within-graph feature variance by Dirichlet energy scaled by the spectral gap; proof by Courant–Fischer per feature dimension (Appendix B).

**Lemma 2** (Lower Lipschitz bound for a GIN block with 2-layer combine, adapted from Chuang and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024). *Let  $h_\theta(\mathbf{X}) = \sigma_2(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$  be a single GIN aggregation  $\mathbf{M} = (1 + \varepsilon)\mathbf{I} + \mathbf{A} \in \mathbb{R}^{N \times N}$  followed by a two-layer MLP with weights  $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$  and  $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$  acting on the feature dimension, with the widening regime  $m \leq m' \leq m''$  and full row rank weights. Assume  $\sigma_{\min}(\mathbf{M}) \geq \kappa > 0$  (e.g.,  $1 + \varepsilon > |\lambda_{\min}(\mathbf{A})|$ ),  $\sigma_{\min}(\mathbf{W}_k) \geq m_k > 0$ , and that  $\sigma_1, \sigma_2$  are leaky-ReLU with slope  $\alpha > 0$ . Then*

$$\|h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F \geq c_{\text{lower}} \|\Delta\mathbf{X}\|_F, \quad c_{\text{lower}} = \alpha^2 m_1 m_2 \kappa. \quad (5)$$

For ordinary ReLU the chain applies locally with  $\alpha = 1$  on segments where all preactivations are strictly positive, giving  $c_{\text{lower}} = m_1 m_2 \kappa$ . A true two-layer GIN  $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$  replaces  $c_{\text{lower}}$  by  $\alpha^2 m_1 m_2 \kappa^2$ .

Proof: chain the global element-wise lower-Lipschitz bound of leaky-ReLU ( $\|\sigma_k(\mathbf{Y} + \Delta) - \sigma_k(\mathbf{Y})\|_F \geq \alpha \|\Delta\|_F$ ) with  $\sigma_{\min}$  of  $\mathbf{W}_2, \mathbf{W}_1, \mathbf{M}$  via  $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$  (Appendix B). The lemma encodes that the GIN does not collapse meaningful feature differences under

injective aggregation and feature transforms, a property the model can exploit when fitting noisy labels.

## 6.2 Main Theorem: Noise Fitting Forces Energy Growth

Consider two graphs  $\mathcal{G}^i, \mathcal{G}^j$  with  $N^i = N^j = N$  and similar pooled descriptors  $\mathbf{g}_i \approx \mathbf{g}_j$  on the clean distribution (same- $N$  for notational simplicity; unequal  $N$  in Appendix B). Suppose label noise flips  $c_j$  to  $c'_j \neq c_i$ , forcing different graph-level outputs. Recall  $\mathbf{Z}^i \in \mathbb{R}^{N \times m'}$  from Section 3; let  $\mathbf{g}_i = \frac{1}{N} \sum_v \mathbf{Z}_v^i$  and  $\tilde{\mathbf{Z}}^i := \mathbf{1}_N \mathbf{g}_i^\top$  (constant-signal replication of  $\mathbf{g}_i$ ).

**Theorem 1** (Graph-level noise fitting forces  $E^{\text{dir}}$  to grow). *Under the hypotheses of Lemmas 1–2 and  $N^i = N^j = N$ ,*

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \leq \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}. \quad (6)$$

In particular, if the noisy training objective combined with an  $L$ -Lipschitz readout forces a graph-level output gap of at least  $\Delta_{\text{out}}$ , so that  $\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \geq \Theta_{\text{noise}} := L^{-1} \Delta_{\text{out}}$ , while the structural similarity gives  $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$ , then

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \geq \Theta_{\text{noise}} - \sqrt{N} \varepsilon. \quad (7)$$

**Reading the bound.** (7) lower-bounds the within-graph Dirichlet energies. The intuition is geometric: for two structurally similar graphs mapped to different labels,  $\mathbf{g}_i \approx \mathbf{g}_j$ , so the required output gap  $\Theta_{\text{noise}}$  cannot come from the means alone; by Poincaré, the only remaining route is the within-graph variance  $E^{\text{dir}}/\lambda_2$ . The unequal- $N$  case (zero-padded matching) is handled in Appendix B. The bound has no node-classification analog: there labels live on the same nodes whose smoothness  $E^{\text{dir}}$  measures, so any link is trivial by construction.

**Empirical anchor.** The bound predicts that  $\sqrt{E^{\text{dir}}/\lambda_2}$  should grow monotonically with the noise rate, since higher noise forces the encoder to discriminate more pairs of similar graphs. The PPA noise-rate sweep in Appendix F confirms this: asymptotic  $E_{\text{set}}^{\text{dir}}$  at 10%, 20%, 30%, 40% noise is monotonically increasing.

## 6.3 Asymmetric Burden on High-Entropy Graphs

(7) only constrains the *sum* of  $\sqrt{E^{\text{dir}}(\mathbf{Z}^i)/\lambda_2(\mathcal{G}^i)}$  across the two graphs. We next show that the structural-entropy ranking of Qian et al. [2025] dictates which graph absorbs the burden.

For a graph  $\mathcal{G}$ , the structural Shannon entropy is  $H(\mathcal{G}) = -\sum_i \frac{d_i}{\sum_j d_j} \log \frac{d_i}{\sum_j d_j}$ . Theorems 1–2 of Qian et al. [2025] prove that more nodes (at fixed sparsity) and denser connectivity (at fixed  $|V|$ ) both strictly increase  $H$ .

**Lemma 3** (Spectral counting). *The combinatorial Laplacian  $\mathbf{L}^i$  has at most  $N^i$  distinct eigenvalues, contained in  $[0, \max_{(u,v) \in \mathcal{E}^i} (d_u + d_v)]$ , hence in  $[0, 2d_{\max}(\mathcal{G}^i)]$  [Chung, 1997]. Equivalently, the normalized spectrum  $\Lambda(\Delta^i) \subset [0, 2]$  with  $\lambda_{\max}(\Delta^i) = 2$  iff  $\mathcal{G}^i$  has a bipartite component. Therefore, when sharpening lifts spectral mass onto eigenvectors  $\bar{\psi}_u$  of  $\bar{\mathbf{L}}$  with  $\bar{\lambda}_u \geq \tau$  for some threshold  $\tau$ , the share allocated to  $\mathcal{G}^i$  is bounded above by the count  $n^i(\tau) = |\{j : \lambda_j^i \geq \tau\}|$ , which is non-decreasing in  $N^i$  and in connection density (by Cauchy interlacing under vertex/edge addition).*

**Corollary 1** (Asymmetric burden). *If  $H(\mathcal{G}^j) \geq H(\mathcal{G}^i)$  via either of the conditions in Qian et al. [2025, Thm. 1–2], then the count of high-frequency eigenvalues of  $\mathcal{G}^j$  above any threshold  $\tau$  is at least that of  $\mathcal{G}^i$ , so the energy increase forced by Theorem 1 is preferentially absorbed by  $\mathcal{G}^j$ . This collapses the node-distribution distinctness of  $\mathcal{G}^j$  [Qian et al., 2025, Sec. 3.3], the very property pooled graph classifiers rely on.*

**Implication.** Theorem 1 says noise fitting must raise energy somewhere; Corollary 1 says it raises it preferentially in the graphs the classifier most needs to distinguish. Suppressing within-graph energy growth therefore blocks the noise-fitting path while preserving the classification signal, an alignment that has no node-classification analog. This also explains why the surge of  $E^{\text{dir}}(Z_2)$  under

noise (Section 5) is steeper than under clean overfitting: noise fitting on similar graphs is structurally constrained to high-frequency, high-entropy directions, while clean overfitting can spread across the full spectrum.

## 7 Three Methods that Reduce Within-Graph Energy

The diagnostic and analytical results above suggest controlling within-graph Dirichlet energy through any of three mechanistically distinct levers: a regularizer on the energy itself, a loss that down-weights samples driving the energy up, or an architectural projection that removes the spectral directions responsible for sharpening. We instantiate one intervention per lever. GCOD is our practical recommendation; CE+W2 is included as a theoretical instantiation.

### 7.1 Explicit Dirichlet-Energy Regularization ( $\mathcal{L}_{DE}$ )

The most direct lever is to penalize  $E^{\text{dir}}(\mathbf{Z}^i)$  above a class-aware threshold  $U_{c_i}$ :

$$\mathcal{L}_{DE}(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^N [\max(0, E^{\text{dir}}(\mathbf{Z}^i) - U_{c_i})]^2, \quad \mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{DE}. \quad (8)$$

We instantiate the threshold in two ways. **Class-specific:**  $U_c$  is recomputed every epoch as the mean  $E^{\text{dir}}$  over a small clean validation subset of class  $c$ . This adapts to class-dependent baseline energies (a molecular graph’s natural energy differs from a social graph’s), but requires clean held-out data. **Fixed:**  $U_c = U$  globally, a single hyperparameter, no clean validation needed, at the cost of mild over-smoothing on clean labels. Both variants improve robustness; class-specific additionally improves *clean-label* accuracy by acting as a generalization-aware regularizer.

### 7.2 Graph Centroid Outlier Discounting (GCOD)

GCOD is our practical recommendation. It extends NCOD [Wani et al., 2023] (originally an image-classification loss) to graph classification, with two design changes that are necessary in the graph-classification regime: a third KL term that disentangles clean from noisy graphs via per-sample trainable parameters  $\mathbf{u} \in [0, 1]^n$ , and feedback through the current training accuracy  $a_{\text{train}}$  that prevents premature outlier discounting. For batch  $B$  with predictions  $f_{\theta}(\mathbf{Z}_B)$ , hard predictions  $\hat{\mathbf{y}}_B$ , soft labels  $\tilde{\mathbf{y}}_B$  (centroid-based, see Appendix C.6), and observed labels  $\mathbf{y}_B$ :

$$\mathcal{L}_1 = \mathcal{L}_{CE}(f_{\theta}(\mathbf{Z}_B) + a_{\text{train}} \text{diag}_{\text{vec}}(\mathbf{u}_B) \mathbf{y}_B, \tilde{\mathbf{y}}_B), \quad (9)$$

$$\mathcal{L}_2 = \frac{1}{|C|} \|\hat{\mathbf{y}}_B + \text{diag}_{\text{vec}}(\mathbf{u}_B) \mathbf{y}_B - \mathbf{y}_B\|^2, \quad (10)$$

$$\mathcal{L}_3 = (1 - a_{\text{train}}) D_{KL}\{\mathcal{L}, \sigma(-\log \mathbf{u}_B)\}, \quad (11)$$

with  $\mathcal{L} = \log \sigma(\text{diag}_{\text{mat}}(f_{\theta}(\mathbf{Z}_B) \mathbf{y}_B^{\top}))$ . Updates are split:  $\theta \leftarrow \theta - \alpha \nabla_{\theta}(\mathcal{L}_1 + \mathcal{L}_3)$ ,  $\mathbf{u} \leftarrow \mathbf{u} - \beta \nabla_{\mathbf{u}} \mathcal{L}_2$ . GCOD is not a theoretically derived energy minimizer. Empirically, however, it reduces  $E_{\text{set}}^{\text{dir}}$  during training (Fig. 2) without explicitly penalizing it: by down-weighting graphs flagged as noisy via large  $u_i$ , it removes the gradient pressure that would otherwise push the network toward high-frequency directions, exactly as Theorem 1 predicts for any sample-level mechanism that suppresses noise-driven gradients.

### 7.3 Spectral Constraint on Weight Matrices (CE+W2)

A third lever, included as a theoretical instantiation, acts directly on the weights. Giovanni et al. [2023] show that for linear graph convolutions with symmetric  $\mathbf{W}$ , the gradient flow of  $E^{\text{dir}}$  decomposes along eigenvectors of  $\mathbf{W}$ : positive eigenvalues smooth, negative ones sharpen. Practical  $\mathbf{W}_l^2$  is not symmetric, so we split  $\mathbf{W}_l^2 = \mathbf{S}_l + \mathbf{N}_l$  and project only  $\mathbf{S}_l$  to its positive eigenspace, leaving the skew  $\mathbf{N}_l$  (purely imaginary spectrum, rotational) untouched. This is a heuristic extension of GRAFF — the energy of  $\mathbf{MZW}$  depends on  $\mathbf{W}^{\top} \mathbf{W}$ , not just  $\mathbf{S}_l$  — but the resulting CE+W2 needs neither clean validation nor extra hyperparameters and shows the energy mechanism can be controlled architecturally. Implementation details are in Appendix C.4; the per-epoch eigendecomposition is costly and can destabilize convergence, so we recommend  $\mathcal{L}_{DE}$  or GCOD for practical use.

Table 1: Test accuracy on the PPA-30%-6 subset (5-layer GIN, mean $\pm$ std over 4 runs). **Best** is highlighted in red and bold, second-best in blue. “CE clean only” shows accuracy on the clean subset of the noisy data, as a reference upper bound. Class-specific  $\mathcal{L}_{DE}^*$  requires a clean validation set.

| Noise | Method                             | Test Acc.                          |              | Train Acc.   |              |
|-------|------------------------------------|------------------------------------|--------------|--------------|--------------|
|       |                                    | Best                               | Last         | Best         | Last         |
| 0%    | CE                                 | 96.25 $\pm$ 0.05                   | 91.25        | 100.0        | 99.33        |
|       | GCOD                               | 96.65 $\pm$ 0.52                   | 93.25        | 99.23        | 99.03        |
|       | CE+W2                              | 96.50 $\pm$ 1.12                   | 86.50        | 99.76        | 99.29        |
|       | $\mathcal{L}_{DE}$ (Fixed)         | 96.58 $\pm$ 0.27                   | 85.70        | 99.26        | 98.29        |
|       | $\mathcal{L}_{DE}^*$ (Class-spec.) | 96.96 $\pm$ 0.04                   | 88.67        | 99.41        | 98.67        |
| 20%   | CE (clean only)                    | 96.15 $\pm$ 0.09                   | 90.66        | 84.50        | 84.07        |
|       | CE                                 | 88.66 $\pm$ 0.16                   | 62.58        | 94.98        | 93.45        |
|       | SOP                                | 91.01 $\pm$ 0.44                   | 85.50        | 79.17        | 77.64        |
|       | <b>GCOD</b>                        | <b>93.91 <math>\pm</math> 0.26</b> | <b>92.58</b> | <b>80.11</b> | <b>79.52</b> |
|       | CE+W2                              | 89.83 $\pm$ 1.23                   | 76.66        | 84.90        | 83.68        |
|       | $\mathcal{L}_{DE}$ (Fixed)         | 89.69 $\pm$ 0.57                   | 80.25        | 78.07        | 70.14        |
|       | $\mathcal{L}_{DE}^*$               | <b>92.34 <math>\pm</math> 0.41</b> | <b>80.92</b> | <b>84.55</b> | <b>83.73</b> |
| 40%   | CE (clean only)                    | 95.08 $\pm$ 0.13                   | 80.44        | 68.15        | 67.05        |
|       | CE                                 | 82.08 $\pm$ 0.22                   | 51.83        | 82.47        | 78.88        |
|       | SOP                                | 82.33 $\pm$ 1.32                   | 65.41        | 58.90        | 57.68        |
|       | <b>GCOD</b>                        | <b>93.88 <math>\pm</math> 0.04</b> | <b>91.08</b> | <b>65.09</b> | <b>64.31</b> |
|       | CE+W2                              | 88.25 $\pm$ 1.13                   | 56.33        | 68.78        | 65.88        |
|       | $\mathcal{L}_{DE}$ (Fixed)         | <b>88.58 <math>\pm</math> 0.75</b> | <b>77.12</b> | <b>61.08</b> | <b>54.53</b> |
|       | $\mathcal{L}_{DE}^*$               | 88.55 $\pm$ 0.11                   | 71.83        | 68.98        | 67.80        |

## 8 Experiments

**Setup.** We evaluate on ogbg-ppa [Szklarczyk et al., 2018a] (full and a controlled 30%/6-class subset), MUTAG, PROTEINS, ENZYMES, IMDB-BINARY, REDDIT-MULTI-12K, MSRC-21, MNIST-graph, and Cora (node classification, transfer test).<sup>1</sup> Backbones are GIN [Xu et al., 2019] and GCN [Kipf and Welling, 2017]; full hyperparameters are in Appendix C. We compare against three baselines: standard cross-entropy (CE); SOP [Liu et al., 2022], a strong noise-robust loss from image classification; and OMG [Yin et al., 2023], the leading graph-classification noise method. All numbers are mean $\pm$ std over 4 runs.

**Why a controlled PPA subset?** On the full 37-class PPA, all methods are under-parameterized for the task, and CE itself is robust (Section 4); a noise-robustness method cannot help where the baseline does not fail. We use a 30%/6-class PPA subset as a *controlled* testbed where over-parameterization triggers noise fitting, and we additionally report the full-PPA result to show our methods do no harm there. Generality claims are grounded in the seven additional benchmarks.

### 8.1 Main Result: Controlled PPA Subset

Table 1 reports the full ablation on the controlled PPA subset. GCOD is best at every noise level and, crucially, has the smallest gap between best and final test accuracy, indicating that it does not overfit late in training. CE+W2 and  $\mathcal{L}_{DE}$  both improve robustness over CE, confirming the convergent-evidence prediction of Section 6. Under clean labels,  $\mathcal{L}_{DE}^*$  even outperforms CE, a direct effect of the class-aware energy threshold.

<sup>1</sup>Anonymized code: [https://anonymous.4open.science/r/Robustness\\_Graph\\_Classification-E76F/](https://anonymous.4open.science/r/Robustness_Graph_Classification-E76F/).



Table 2: GCOD matches or beats CE across diverse datasets. Mean over 4 runs;  $\pm$ std reported where available. *Sym*<sub>20</sub>: 20% symmetric noise. *Asym*<sub>40</sub>: 40% asymmetric noise. The *Asym*<sub>40</sub> GCOD column reports the std deviations from [Wani et al., 2023]’s rebuttal protocol.

| Dataset  | 0% CE | Sym <sub>20</sub> CE | Sym <sub>20</sub> GCOD | Asym <sub>40</sub> CE | Asym <sub>40</sub> GCOD |
|----------|-------|----------------------|------------------------|-----------------------|-------------------------|
| PROTEINS | 81.16 | 76.23                | 79.38                  | 72.90                 | 76.19 $\pm$ 1.22        |
| MNIST    | 72.69 | 66.30                | 71.26                  | 71.15                 | 72.61 $\pm$ 0.43        |
| ENZYMES  | 73.33 | 64.16                | 68.69                  | 65.80                 | 69.81 $\pm$ 0.98        |
| IMDB-B   | 76.50 | 75.00                | 75.40                  | 68.18                 | 72.89 $\pm$ 0.91        |
| MUTAG    | 94.73 | 89.47                | 90.01                  | 91.16                 | 93.19 $\pm$ 0.38        |
| REDDIT   | 48.15 | 45.05                | 44.98                  | 48.01                 | 47.94 $\pm$ 2.87        |
| MSRC-21  | 96.69 | 90.26                | 94.69                  | 94.39                 | 95.57 $\pm$ 0.32        |

Table 3: OMG protocol (Yin et al., 2023). Percentage gain over CE.

| Dataset  | OMG   | GCOD  |
|----------|-------|-------|
| MUTAG    | 0.061 | 0.062 |
| IMDB-B   | 0.047 | 0.049 |
| PROTEINS | 0.039 | 0.041 |

Table 4: Relative training time (CE-GIN = 1.00).

| Method                 | Runtime        |
|------------------------|----------------|
| CE-GIN                 | 1.00           |
| SOP                    | 1.048          |
| GCOD                   | <b>1.029</b>   |
| CE+W2                  | 1.33           |
| OMG [Yin et al., 2023] | $\approx$ 3.00 |

## 8.2 Generalization Across Datasets and Noise Types

Table 2 reports GCOD across seven benchmarks beyond PPA. GCOD consistently improves over CE under both 20% symmetric and 40% asymmetric noise, and matches CE on REDDIT where the baseline is already saturated by class imbalance. The empirical trend that gains grow with the noise rate (Appendix F) is consistent with Theorem 1: the forced energy lower bound  $\Theta_{\text{noise}} - \sqrt{N}\varepsilon$  scales with the noise level, so the room a method has to help also scales.

**Full ogbg-ppa.** Under the full 37-class benchmark, all methods including CE behave robustly to noise (Section 4); we therefore use it only as a do-no-harm check. Detailed results are in Appendix H.

**Comparison with OMG.** Under the OMG [Yin et al., 2023] protocol (Table 3), GCOD matches OMG’s accuracy (within 0.1–0.2 pp) while requiring  $\sim 70\times$  less training overhead (Table 4): OMG adds  $\approx 200\%$  runtime, GCOD only  $\approx 2.9\%$ .

**Cross-setting transfer to node classification (Cora).** Applied unchanged to Cora under three structured 20% noise types (flip, random, uniform) against 13 node-specific baselines (full table in Appendix I), our methods rank in the top three on every backbone-noise combination. Highlights: on GAT, GCOD reaches 0.787 (#2) under uniform noise and CE+W2 0.8157 (#2) under clean labels; on GCN, CE+W2 reaches 0.7743 (#3) under random noise and GCOD 0.7687 (#3) under flip noise. That an energy-based regularizer using neither neighbor labels nor homophily matches specialized node-classification methods is direct evidence the perspective captures a general property of GNN training under label noise.

**Convergent evidence.** The three methods (architectural, regularization, sample re-weighting) all reduce  $E_{\text{set}}^{\text{dir}}$  during training and improve robustness over CE — convergent evidence for the energy mechanism (Fig. 2, Appendix F).

## 9 Conclusion

Within-graph Dirichlet energy diagnoses graph-level noise memorization, supported by a Poincaré-based bound and an entropy asymmetry argument unique to graph classification. Three interventions ( $\mathcal{L}_{DE}$ , GCOD, CE+W2) reduce energy and improve robustness on eight benchmarks; GCOD matches OMG at  $\sim 70\times$  lower compute. **Limitations:** sufficient bound only, synthetic noise, heuristic CE+W2. **Impact:** robust GNNs aid chemistry/biology; GCOD needs no clean validation.

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## A Additional Related Work

### A.1 Learning under Label Noise

Beyond robust losses, additional families include sample relabelling [Arazo et al., 2019, Reed et al., 2015], two-network co-teaching and divide-and-mix approaches [Han et al., 2018, Li et al., 2020, Kim et al., 2023], and regularization through MixUp [Zhang et al., 2017]. Reweighting methods adapt sample importance via training dynamics or held-out probes [Liu and Tao, 2015, Pleiss et al., 2020]. Most of these were developed for image classification; their direct transfer to graph classification is non-obvious because graphs lack the strong feature regularities of images.

### A.2 Graph Learning under Noise

Most graph-related noise work targets node classification, exploiting either homophily (RS-GNN [Dai et al., 2022], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a], pairwise interactions [Du et al., 2023]) or self-supervision (pseudo-label denoising [Yuan et al., 2023b, Qian et al., 2023]). Edge/feature noise has also been studied, e.g. structural noise [Fox and Rajamanickam, 2019, Dai et al., 2022]. These methods typically assume connected nodes share labels, which is incompatible with our graph-classification setting where labels are graph-level. The seminal work of NT et al. [2019] treats graph classification under label noise via a surrogate loss; OMG [Yin et al., 2023] combines contrastive learning, MixUp, and curriculum filtering. Both are practical improvements but provide no spectral or energy-based explanation of *why* GNNs fit graph-level noise.

### A.3 Dirichlet Energy and Spectral Properties of GNNs

NT and Maehara [2019] and Rusch et al. [2023] analyze GNNs as low-pass filters, with self-loops further shrinking eigenvalues. Cai and Wang [2020], Oono and Suzuki [2020] prove that GNN Dirichlet energy decays exponentially when the product of the largest singular value of the weight matrix and the largest Laplacian eigenvalue is below 1. Giovanni et al. [2023] disentangle smoothing/sharpening via positive/negative weight-eigenvalues. Zhou et al. [2021] (NeurIPS) uses energy regularization to prevent over-smoothing on *node* classification. Our setting is the opposite regime: too *much* energy under graph-level label noise.

Qian et al. [2025] introduce a structural-entropy viewpoint on over-smoothing for graph classification, showing that high-entropy regions are most damaged by over-smoothing. We invert and combine this with Proposition 1 to get Corollary 1: the graphs that suffer most from over-*sharpening* (the noise-fitting regime) are also the high-entropy graphs that carry the most discriminative signal. The two papers therefore identify symmetric ends of the same spectral failure.

### A.4 Lipschitz/Stability Analyses

Chuang and Jegelka [2022] bound GIN outputs via the Tree Mover’s Distance, and Davidson and Dym [2025] extend uniform-Lipschitz analyses to MPNNs through Hölder continuity of aggregation/combination/readout. Juvina et al. [2024] derive optimal Lipschitz constants  $\vartheta = \phi(\lambda_K)$  for graph convolutions under non-negative weights and ReLU. Gama et al. [2019] show output stability under graph-shift-operator perturbations. These results characterize stability w.r.t. inputs; we instead study stability against *label* noise, mediated by the within-graph energy of learned representations.

## B Proofs and Extended Theory

### B.1 Proof of Proposition 1

Let  $\Lambda^i = \{\lambda_u^i\}_{u=1}^{N^i}$  be the spectrum of the per-graph combinatorial Laplacian  $\mathbf{L}^i = \mathbf{D}^i - \mathbf{A}^i$ . Define the block-diagonal Laplacian

$$\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n), \quad \det(\bar{\mathbf{L}} - \lambda \mathbf{I}_{N_{\text{tot}}}) = \prod_{i=1}^n \det(\mathbf{L}^i - \lambda \mathbf{I}_{N^i}), \quad (12)$$

so  $\bar{\Lambda} = \bigcup_i \Lambda^i$ . Let  $\psi_j^i$  be the  $j$ -th eigenvector of  $\mathbf{L}^i$  at eigenvalue  $\lambda_j^i$ . The vector

$$\bar{\psi}_u = (0, \dots, 0, \psi_j^i, 0, \dots, 0)^\top \in \mathbb{R}^{N_{\text{tot}}} \quad (13)$$

(zero on every block other than  $i$ ) satisfies  $\bar{\mathbf{L}}\bar{\psi}_u = \lambda_j^i \bar{\psi}_u$ . With  $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \parallel \dots \parallel \mathbf{Z}^n]$  (vertical concatenation), the combinatorial Dirichlet energy of the block-diagonal graph satisfies

$$E^{\text{dir}}(\bar{\mathbf{Z}}) = \text{tr}(\bar{\mathbf{Z}}^\top \bar{\mathbf{L}} \bar{\mathbf{Z}}) = \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2 = \sum_i \sum_{r,j} \lambda_j^i (\psi_j^{i\top} \mathbf{Z}_{:,r}^i)^2 = \sum_i E^{\text{dir}}(\mathbf{Z}^i), \quad (14)$$

where the second equality uses the spectral expansion in the orthonormal basis  $\{\bar{\psi}_u\}$  and the third uses the block-supported structure of each  $\bar{\psi}_u$ . Hence  $E_{\text{set}}^{\text{dir}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_i E^{\text{dir}}(\mathbf{Z}^i) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}})$ .  $\square$

## B.2 Proof of Lemma 1 (Graph Poincaré Inequality)

The combinatorial Laplacian  $\mathbf{L} = \mathbf{D} - \mathbf{A}$  is real symmetric and positive semi-definite. Because  $\mathcal{G}$  is connected,  $\ker(\mathbf{L}) = \text{span}(\mathbf{1})$ , and the eigenvalues are  $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$ . By the Courant–Fischer min–max characterization,

$$\lambda_2(\mathcal{G}) = \min_{\substack{\mathbf{x} \perp \mathbf{1} \\ \mathbf{x} \neq 0}} \frac{\mathbf{x}^\top \mathbf{L} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}. \quad (15)$$

Fix any feature dimension  $r$  and let  $\mathbf{z} = \mathbf{Z}_{:,r} \in \mathbb{R}^N$  with  $\bar{z} = \frac{1}{N} \sum_v z_v$ . The centered vector  $\mathbf{z} - \bar{z}\mathbf{1}$  is orthogonal to  $\mathbf{1}$ , so

$$\lambda_2(\mathcal{G}) \|\mathbf{z} - \bar{z}\mathbf{1}\|_2^2 \leq (\mathbf{z} - \bar{z}\mathbf{1})^\top \mathbf{L} (\mathbf{z} - \bar{z}\mathbf{1}) = \mathbf{z}^\top \mathbf{L} \mathbf{z} = \sum_{(u,v) \in \mathcal{E}} (z_u - z_v)^2, \quad (16)$$

using  $\mathbf{L}\mathbf{1} = \mathbf{0}$ . The right-hand side is the per-feature-dimension contribution to  $E^{\text{dir}}(\mathbf{Z})$  in (1). Summing over  $r = 1, \dots, m'$  yields

$$\sum_{v=1}^N \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 = \sum_{r=1}^{m'} \|\mathbf{z}^{(r)} - \bar{z}^{(r)}\mathbf{1}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} \sum_{r=1}^{m'} \mathbf{z}^{(r)\top} \mathbf{L} \mathbf{z}^{(r)} = \frac{E^{\text{dir}}(\mathbf{Z})}{\lambda_2(\mathcal{G})}, \quad (17)$$

which is (4). The constant  $1/\lambda_2$  is sharp, attained by the Fiedler vector  $\psi_2$ . The corresponding bound for the normalized Dirichlet energy  $E_{\text{sym}}^{\text{dir}}(\mathbf{Z}) = \text{tr}(\mathbf{Z}^\top \Delta \mathbf{Z})$  is obtained by the change of variable  $\mathbf{y}_v = \mathbf{Z}_v / \sqrt{d_v}$  and applying the same argument with  $\lambda_2(\mathbf{L})$  in the bound.  $\square$

## B.3 Proof of Lemma 2 (Lower Lipschitz Bound for a GIN)

The block under analysis is one GIN aggregation followed by a two-layer MLP combine,

$$h_\theta(\mathbf{X}) = \sigma_2(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1) \mathbf{W}_2), \quad (18)$$

acting on  $\mathbf{X} \in \mathbb{R}^{N \times m}$  with  $\mathbf{M} \in \mathbb{R}^{N \times N}$  mixing rows, and  $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$ ,  $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$  acting on the feature dimension.

**Global lower-Lipschitz of leaky-ReLU.** For scalar  $a, b \in \mathbb{R}$ ,  $|\text{LReLU}_\alpha(a) - \text{LReLU}_\alpha(b)| \geq \alpha|a - b|$ , since the function has slope  $\geq \alpha > 0$  everywhere. Applied element-wise to a matrix  $\mathbf{Y}$  and a perturbation  $\mathbf{Y} + \Delta \mathbf{Y}$  this gives

$$\|\sigma_k(\mathbf{Y} + \Delta \mathbf{Y}) - \sigma_k(\mathbf{Y})\|_F \geq \alpha \|\Delta \mathbf{Y}\|_F, \quad k = 1, 2. \quad (19)$$

This is a *global* statement, not a local one — no strict-margin or single-region assumption is needed for leaky-ReLU.

**Lower bounds for right- and left-multiplication.** Using  $\text{vec}(\mathbf{A}\mathbf{Y}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A})\text{vec}(\mathbf{Y})$  and  $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$ , for any matrix  $\mathbf{Z}$ ,

$$\|\mathbf{Z}\mathbf{W}_2\|_F = \|(\mathbf{W}_2^\top \otimes \mathbf{I})\text{vec}(\mathbf{Z})\|_2 \geq \sigma_{\min}(\mathbf{W}_2)\|\mathbf{Z}\|_F \geq m_2\|\mathbf{Z}\|_F, \quad (20)$$

$$\|\mathbf{M}\mathbf{Z}\mathbf{W}_1\|_F = \|(\mathbf{W}_1^\top \otimes \mathbf{M})\text{vec}(\mathbf{Z})\|_2 \geq \sigma_{\min}(\mathbf{W}_1)\sigma_{\min}(\mathbf{M})\|\mathbf{Z}\|_F \geq m_1\kappa\|\mathbf{Z}\|_F. \quad (21)$$



574 The widening regime  $m \leq m' \leq m''$  together with  $\sigma_{\min}(\mathbf{W}_k) > 0$ ,  $\sigma_{\min}(\mathbf{M}) > 0$  ensures that each  
 575 Kronecker factor has full column rank, so the lower-bound inequalities above are non-trivial.

576 **Chain.** Apply (19) for  $\sigma_2$ , then right-multiplication by  $\mathbf{W}_2$ , then (19) for  $\sigma_1$ , then  $\mathbf{M} \cdot \mathbf{W}_1$ :

$$\begin{aligned} \|h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F &\geq \alpha \|\sigma_1(\mathbf{M}(\mathbf{X} + \Delta\mathbf{X})\mathbf{W}_1)\mathbf{W}_2 - \sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2\|_F \\ &\geq \alpha m_2 \|\sigma_1(\mathbf{M}(\mathbf{X} + \Delta\mathbf{X})\mathbf{W}_1) - \sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\|_F \\ &\geq \alpha^2 m_2 \|\mathbf{M}\Delta\mathbf{X}\mathbf{W}_1\|_F \\ &\geq \alpha^2 m_1 m_2 \kappa \|\Delta\mathbf{X}\|_F = c_{\text{lower}} \|\Delta\mathbf{X}\|_F. \end{aligned} \quad (22)$$

577 **Equivalent vectorized Jacobian view.** Writing the Jacobian in vec form,  $\mathbf{J} = \mathbf{D}_2(\mathbf{W}_2^\top \otimes$   
 578  $\mathbf{I}_N)\mathbf{D}_1(\mathbf{W}_1^\top \otimes \mathbf{M})$ , with  $\mathbf{D}_k \succeq \alpha \mathbf{I}$  for leaky-ReLU. Using  $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$   
 579 and the singular-value chain  $\sigma_{\min}(\mathbf{ABCD}) \geq \prod \sigma_{\min}$  (valid under full column rank of each factor)  
 580 recovers  $\sigma_{\min}(\mathbf{J}) \geq \alpha^2 m_1 m_2 \kappa$ . The element-wise chain above is what turns this Jacobian bound  
 581 into the global statement (5) for leaky-ReLU.

582 **Two-layer GIN.** For  $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$ , an additional left-multiplication by  $\mathbf{M}$  enters the  
 583 chain, contributing an extra  $\sigma_{\min}(\mathbf{M}) \geq \kappa$  and giving  $c_{\text{lower}} = \alpha^2 m_1 m_2 \kappa^2$ .

584 **Ordinary ReLU.** The element-wise lower-Lipschitz bound (19) fails globally for ReLU: on any  
 585 segment that crosses the origin in some coordinate,  $|\text{ReLU}(a) - \text{ReLU}(b)| < |a - b|$  is possible.  
 586 The chain therefore only works on regions where every coordinate of every preactivation is strictly  
 587 positive along  $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$ . On such a region,  $\sigma_k$  acts as the identity (slope 1), so the chain  
 588 gives  $c_{\text{lower}} = m_1 m_2 \kappa$  (the formula in (5) with  $\alpha = 1$ ).  $\square$

#### 589 B.4 Proof of Theorem 1 (Noise Forces Energy Growth)

590 Let  $\mathbf{Z}^i, \mathbf{Z}^j \in \mathbb{R}^{N \times m'}$  be the post-message-passing per-node representations of  $\mathcal{G}^i$  and  $\mathcal{G}^j$  as defined in  
 591 Section 3 (we assume  $N^i = N^j = N$ ; the unequal- $N$  case is treated below). Let  $\mathbf{g}_i = \frac{1}{N} \sum_v \mathbf{Z}_v^i \in$   
 592  $\mathbb{R}^{m'}$  be the mean-pooled graph descriptor, and let  $\tilde{\mathbf{Z}}^i = \mathbf{1}_N \mathbf{g}_i^\top$  denote its replication to a constant  
 593 per-node signal. By the Frobenius-norm triangle inequality,

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \leq \|\mathbf{Z}^i - \tilde{\mathbf{Z}}^i\|_F + \|\tilde{\mathbf{Z}}^i - \tilde{\mathbf{Z}}^j\|_F + \|\tilde{\mathbf{Z}}^j - \mathbf{Z}^j\|_F. \quad (23)$$

594 **Middle term.** Because both  $\tilde{\mathbf{Z}}^i, \tilde{\mathbf{Z}}^j$  are constant signals of size  $N \times m'$ ,

$$\|\tilde{\mathbf{Z}}^i - \tilde{\mathbf{Z}}^j\|_F^2 = \sum_{v=1}^N \|\mathbf{g}_i - \mathbf{g}_j\|_2^2 = N \|\mathbf{g}_i - \mathbf{g}_j\|_2^2, \quad (24)$$

595 so  $\|\tilde{\mathbf{Z}}^i - \tilde{\mathbf{Z}}^j\|_F = \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2$ .

596 **Variance terms.** Lemma 1 applied to  $\mathbf{Z}^i$  gives

$$\|\mathbf{Z}^i - \tilde{\mathbf{Z}}^i\|_F^2 = \sum_v \|\mathbf{Z}_v^i - \mathbf{g}_i\|_2^2 \leq \frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}, \quad (25)$$

597 and analogously for  $\mathcal{G}^j$ . Taking square roots and combining,

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \leq \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}, \quad (26)$$

598 which is (6).

599 **Lower bound from the noisy training objective.** The graph-level prediction is  $\hat{\mathbf{y}}_i = f_\theta(\mathbf{Z}^i)$ , where  
 600  $f_\theta$  is the permutation-invariant readout from Section 3; we assume  $f_\theta$  is  $L$ -Lipschitz in the Frobenius  
 601 norm (mean- or sum-pooling followed by an MLP with bounded weights). At convergence, the noisy  
 602 training objective enforces  $\hat{\mathbf{y}}_i$  to lie near a one-hot vector at  $c_i$  and  $\hat{\mathbf{y}}_j$  near a one-hot at  $c_j \neq c_i$ , so  
 603  $\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq \Delta_{\text{out}}$  for some  $\Delta_{\text{out}} > 0$ . By the readout's upper-Lipschitz property,

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \geq L^{-1} \|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq L^{-1} \Delta_{\text{out}} =: \Theta_{\text{noise}}. \quad (27)$$

604 The role of Lemma 2 is to make this lower bound non-vacuous:  $c_{\text{lower}} > 0$  ensures that the encoder has  
 605 a positive lower-Lipschitz constant at convergence, so the noisy-label constraint cannot be satisfied

by collapsing  $(\mathbf{X}^i, \mathbf{A}^i)$  and  $(\mathbf{X}^j, \mathbf{A}^j)$  to the same internal representation; equivalently, the encoder remains in a regularity regime where  $\|\mathbf{Z}^i - \mathbf{Z}^j\|_F$  is a meaningful quantity.

**Combining upper and lower bounds.** Under the structural-similarity assumption  $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$ , substituting  $\Theta_{\text{noise}} \leq \|\mathbf{Z}^i - \mathbf{Z}^j\|_F$  into (6) and rearranging,

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \geq \Theta_{\text{noise}} - \sqrt{N} \varepsilon, \quad (28)$$

which is (7).  $\square$

**Unequal graph sizes.** When  $N^i \neq N^j$ , set  $N = \max(N^i, N^j)$  and pad the smaller graph by appending  $N - N^{\min}$  rows equal to its mean  $\mathbf{g}_\bullet$ , with the corresponding padded vertices isolated (no edges). Let  $(\mathbf{Z}^\bullet)^{\text{pad}}$  and  $(\tilde{\mathbf{Z}}^\bullet)^{\text{pad}}$  denote the padded representation and its constant-signal replication. Two facts make this padding harmless: (i) appending  $\mathbf{g}_\bullet$ -valued rows leaves the graph mean unchanged, and (ii) isolated padded vertices contribute nothing to the Dirichlet energy, so  $E^{\text{dir}}((\mathbf{Z}^\bullet)^{\text{pad}}) = E^{\text{dir}}(\mathbf{Z}^\bullet)$ . The within-graph variance  $\|(\mathbf{Z}^\bullet)^{\text{pad}} - (\tilde{\mathbf{Z}}^\bullet)^{\text{pad}}\|_F^2 = \sum_{v=1}^{N^{\min}} \|\mathbf{Z}_v^\bullet - \mathbf{g}_\bullet\|^2$  is also preserved, so Lemma 1 applies unchanged. The middle term becomes  $\|(\mathbf{Z}^i)^{\text{pad}} - (\tilde{\mathbf{Z}}^j)^{\text{pad}}\|_F = \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2$ , and the readout’s lower bound is preserved up to a fixed pooling constant. Hence (6) and (7) hold with  $N \rightarrow \max(N^i, N^j)$ .

## B.5 Proof of Lemma 3 (Spectral Counting)

The combinatorial Laplacian  $\mathbf{L}^i$  is symmetric positive semi-definite. By Anderson–Morley’s bound, every eigenvalue  $\lambda_j^i$  of  $\mathbf{L}^i$  satisfies  $\lambda_j^i \leq \max_{(u,v) \in \mathcal{E}^i} (d_u + d_v) \leq 2d_{\max}(\mathcal{G}^i)$ . Equivalently, the normalized Laplacian  $\Delta^i = \mathbf{D}^{-1/2} \mathbf{L}^i \mathbf{D}^{-1/2}$  has eigenvalues in  $[0, 2]$ , with  $\lambda_{\max}(\Delta^i) = 2$  iff  $\mathcal{G}^i$  has a bipartite component [Chung, 1997, Lemma 1.7, Thm. 1.13]. The number of distinct eigenvalues of  $\mathbf{L}^i$  is at most  $N^i$ .

For any threshold  $\tau > 0$ , define  $n^i(\tau) = |\{j : \lambda_j^i \geq \tau\}|$ . We claim  $n^i(\tau)$  is non-decreasing under (a) edge addition and (b) vertex addition with at least one incident edge.

(a) *Edge addition:* if  $\mathcal{G}^{i'} = \mathcal{G}^i + \{(u, v)\}$ , then  $\mathbf{L}^{i'} = \mathbf{L}^i + (\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top$ , a rank-one PSD update. By Weyl’s monotonicity theorem (Horn and Johnson, 2012, Cor. 4.3.12),  $\lambda_k(\mathbf{L}^{i'}) \geq \lambda_k(\mathbf{L}^i)$  for every  $k$ . Hence if  $\lambda_j^i \geq \tau$  then  $\lambda_j^{i'} \geq \tau$ , so  $n^{i'}(\tau) \geq n^i(\tau)$ .

(b) *Vertex addition:* write  $\mathcal{G}^{i'}$  as the result of starting from the padded graph  $\mathcal{G}^i \sqcup \{v_{N+1}\}$  (an isolated extra vertex) and then adding all incident edges  $\{(v_{N+1}, u) : u \in \mathcal{N}(v_{N+1})\}$  one at a time. The padded Laplacian is  $\mathbf{L}_0 = \mathbf{L}^i \oplus [0] \in \mathbb{R}^{(N^i+1) \times (N^i+1)}$ , with spectrum  $\Lambda(\mathbf{L}^i) \cup \{0\}$ . Each subsequent edge addition is a rank-one PSD update covered by case (a), so  $\lambda_k(\mathbf{L}^{i'}) \geq \lambda_k(\mathbf{L}_0)$  for all  $k$ . Since  $\Lambda(\mathbf{L}_0) \setminus \{0\} = \Lambda(\mathbf{L}^i) \setminus \{0\}$ , the count above any  $\tau > 0$  satisfies  $n^{i'}(\tau) \geq n_0(\tau) = n^i(\tau)$ . (The case  $\tau = 0$  is trivial since  $n^{i'}(0) = N^i + 1 \geq N^i = n^i(0)$ .)

Now let  $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$  as in Proposition 1. By the block-diagonal structure, every eigenpair  $(\bar{\lambda}_u, \bar{\psi}_u)$  of  $\bar{\mathbf{L}}$  is supported on exactly one  $\mathcal{G}^i$ . Sharpening that lifts spectral mass onto eigenvectors with  $\bar{\lambda}_u \geq \tau$  therefore allocates a share to  $\mathcal{G}^i$  that is bounded above by  $n^i(\tau) / \sum_k n^k(\tau)$ , which grows with  $N^i$  and connection density of  $\mathcal{G}^i$ .  $\square$

## B.6 Proof of Corollary 1 (Asymmetric Burden)

Suppose  $H(\mathcal{G}^j) \geq H(\mathcal{G}^i)$  either via more nodes at fixed sparsity (Theorem 1 of Qian et al., 2025) or via denser connectivity at fixed  $|V|$  (Theorem 2 of Qian et al., 2025). Lemma 3 then implies  $n^j(\tau) \geq n^i(\tau)$  for every  $\tau$ , where  $n^i(\tau)$  counts eigenvalues of  $\mathcal{G}^i$  above  $\tau$ . Now consider any sharpening update of  $\bar{\mathbf{Z}}$  that increases the spectral coefficients on eigenpairs with  $\bar{\lambda}_u \geq \tau$ . By Proposition 1, each such eigenpair lies entirely on one of  $\mathcal{G}^i$  or  $\mathcal{G}^j$ , and the count of available high-frequency eigenvectors on  $\mathcal{G}^j$  is at least that on  $\mathcal{G}^i$ . Hence on average a  $\bar{\lambda}_u \geq \tau$  component is more likely to be supported on  $\mathcal{G}^j$ , and the resulting energy increase is preferentially absorbed by  $\mathcal{G}^j$ . By Section 3.3 of Qian et al. [2025], increasing  $E^{\text{dir}}(\mathbf{Z}^j)$  along its high-frequency eigenvectors collapses the pairwise distinctness of node-feature distributions inside  $\mathcal{G}^j$ , the precise quantity that pooled graph classifiers rely on for separation. Hence high-entropy graphs lose discriminability faster than low-entropy graphs.  $\square$

## 653 B.7 Why a Decrease of $E_{\text{set}}^{\text{dir}}$ Is Non-Trivial in Graph Classification

654 Three consecutive observations together justify the term “non-trivial”:

655 (i) Noise is injected at the graph-label level, not at the node-feature level. Therefore, the increase  
656 of  $E^{\text{dir}}(\mathbf{Z}^i)$  during noise fitting cannot be a re-encoding of the noise itself; it must be a behavioral  
657 response of the network through training dynamics.

658 (ii) For a randomly-initialized network,  $E_{\text{set}}^{\text{dir}}$  is independent of class labels by construction. Any  
659 correlation between  $E_{\text{set}}^{\text{dir}}$  and labels emerges from the optimization process, not from the data.

660 (iii) By Theorem 1 and Corollary 1, the only optimization path that fits incorrect graph-level labels is  
661 to lift spectral mass into high-entropy directions of  $\bar{\mathbf{L}}$ , which are precisely the directions that a clean,  
662 generalization-driven network should preserve. Constraining  $E_{\text{set}}^{\text{dir}}$  therefore selectively blocks the  
663 noise-fitting path while allowing the clean-learning path, an alignment that does *not* hold for node  
664 classification, where label noise corrupts the same nodes whose smoothness the energy measures,  
665 and where decreasing energy can erase the discriminative signal.

666 These three points collectively make the connection between energy reduction and noise robustness  
667 in graph classification a non-trivial property. The analogous statement in node classification reduces  
668 to “the model fits homophily, and homophily is what we measured.”

## 669 C Experimental Setting

### 670 C.1 Datasets

671 **ogbg-ppa** [Szkarczyk et al., 2018a]: 158,100 protein-association graphs across 37 taxonomic groups,  
672 derived from STRING [Szkarczyk et al., 2018b]; the standard OGB graph-property-prediction  
673 benchmark. Modified PPA-30%-6 takes 30% of training graphs from 6 selected classes, used as a  
674 controlled testbed. **PROTEINS** [Borgwardt et al., 2005]: 1,113 protein graphs; 2 classes; mean 39  
675 nodes/graph. **ENZYMES** [Borgwardt et al., 2005]: 600 enzyme graphs across 6 EC top-level classes.  
676 **MSRC-21** [Neumann et al., 2016]: 563 image-segmentation graphs, 20 classes. **MUTAG** [Kriege  
677 and Mutzel, 2012]: 188 molecular graphs, 2 classes. **IMDB-BINARY** [Yanardag and Vishwanathan,  
678 2015b]: 1,000 social graphs, 2 classes. **REDDIT-MULTI-12K** [Yanardag and Vishwanathan, 2015b]:  
679 11,929 thread-graphs, 11 classes. **MNIST-graph**: 55,000 superpixel graphs over the MNIST images,  
680 10 classes.

### 681 C.2 Synthetic Datasets for Failure-Mode F2

682 We sample average graph orders  $\bar{N} \in \{5, 10, \dots, 60\}$ , drawing actual node counts from  $\text{Poisson}(\bar{N})$ .  
683 Each dataset has 6 classes with 1,400 graphs/class, fixed average degree 2, edges sampled uniformly.  
684 Node features are  $\mathcal{N}(\mu_c, 1.5)$  with  $\mu_c = f(\text{class})$ . Symmetric noise rate is fixed at 35%.

### 685 C.3 Architectures and Hyperparameters

686 Unless otherwise stated, we use a 5-layer GIN [Xu et al., 2019] or GCN [Kipf and Welling, 2017]  
687 with 300 hidden units, Adam ( $\eta = 10^{-3}$ , weight decay  $10^{-4}$ ), batch size 32, 200–1000 epochs,  
688  $\eta_u = 1$  for GCOD’s outlier parameter. Compute: NVIDIA RTX A6000.

### 689 C.4 Spectral Bias Implementation Details (CE+W2)

690 For each layer  $l$ , after a gradient update, decompose  $\mathbf{W}_l^2 = \Phi_l^2 \mu_l^2 (\Phi_l^2)^{-1}$  and replace by  
691  $\Phi_l^2 [\mu_l^2]^+ (\Phi_l^2)^{-1}$ , where  $[\cdot]^+$  is element-wise ReLU on the eigenvalues. The projection is detached  
692 from the gradient graph. We apply it to the post-aggregation matrix  $\mathbf{W}^2$  of every layer; applying it  
693 also to  $\mathbf{W}^1$  marginally improves smoothing but degrades stability. Convergence variance under this  
694 scheme is shown in Fig. 8.

### 695 C.5 Class-Specific and Fixed Bounds for $\mathcal{L}_{DE}$

696 **Class-specific:** after every epoch, set  $U_c = |\mathcal{D}_c^{\text{val}}|^{-1} \sum_{i \in \mathcal{D}_c^{\text{val}}} E^{\text{dir}}(\mathbf{Z}^i)$ . The validation set must be  
 697 clean to keep  $U_c$  class-specific. **Fixed:**  $U_c = U$  globally. We progressively decrease  $U$  from a large  
 698 initial value until accuracy under clean labels begins to drop; the final  $U$  is the largest value that  
 699 achieves a noticeable energy ceiling without over-smoothing.

### 700 C.6 Design of GCOD

701 We adopt the centroid-based soft labels of Wani et al. [2023]: normalize  $\phi(\mathbf{z}_i) = \mathbf{z}_i / \|\mathbf{z}_i\|$ ; per-  
 702 class centroid  $\phi_c = \bar{\mathbf{z}}_c / \|\bar{\mathbf{z}}_c\|$ ; soft target  $\tilde{\mathbf{y}}_i = [\max(\phi(\mathbf{z}_i)\phi_c^\top, 0)\mathbf{y}_i]$ . Equations (9)–(11) are then  
 703 optimized as in the main text. The training-accuracy feedback in  $\mathcal{L}_1$  and  $\mathcal{L}_3$  is critical: without it,  $\mathbf{u}$   
 704 dominates early in training and discounts samples before genuine patterns are learned (Fig. 7).

## 705 D Failure-Mode Experiments (Extended)

706 Additional plots for the F1–F3 failure modes are provided here for completeness, including effect-of-  
 707 dataset-size for both PPA and synthetic datasets.

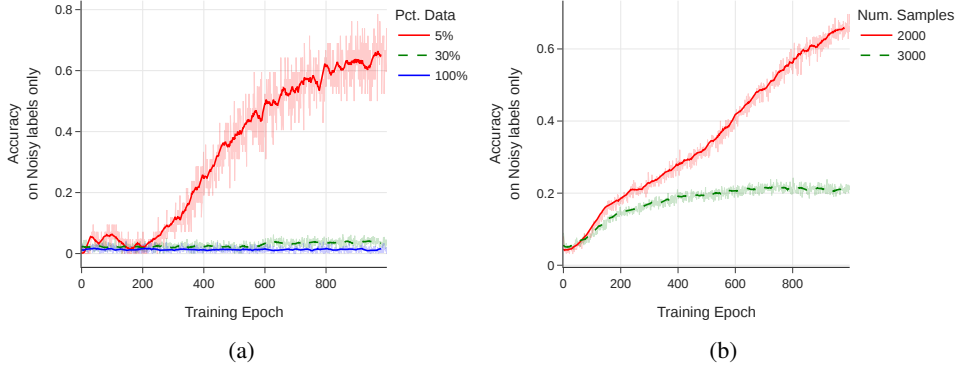


Figure 3: Effect of training-set size on noise memorization (training accuracy on noisy labels only).  
 (a) PPA subsampling at 20% symmetric noise. (b) Synthetic graphs of order 7, 6 classes, 35% noise.

## 708 E Dirichlet-Energy Amplification in High-Frequency Components

709 We use the HLFF-GNN framework [Xu et al., 2024] (implemented as FGRLConv) to decompose  
 710 representations into low-frequency  $Z_1$ , high-frequency  $Z_2$ , and residual  $Y$ . Under the composite loss

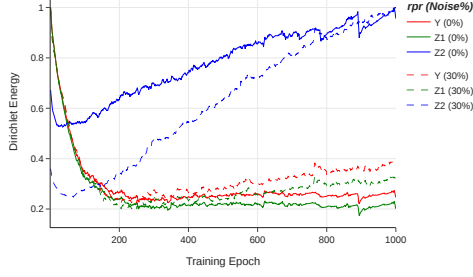
$$\mathcal{L} = \|Y - X\|_F^2 + \lambda \text{tr}(Z_1^\top \Delta Z_1) + \alpha \text{tr}(Z_2^\top (\mathbf{I} - \Delta) Z_2) + \beta (\|Y^\top Z_1\|_F^2 + \|Y^\top Z_2\|_F^2), \quad (29)$$

711 the spectral form of the high-frequency energy is  $E^{\text{dir}}(Z_2) = \sum_{r,u} \lambda_u (\psi_u^\top Z_{2,r})^2$ , in which large  $\lambda_u$   
 712 amplify any noise-driven content. On ENZYMES with 30% symmetric noise, we observe a sharp  
 713 monotone increase in  $E^{\text{dir}}(Z_2)$ , while  $E^{\text{dir}}(Z_1)$  and  $E^{\text{dir}}(Y)$  remain stable, consistent with Theorem 1  
 714 and Corollary 1.

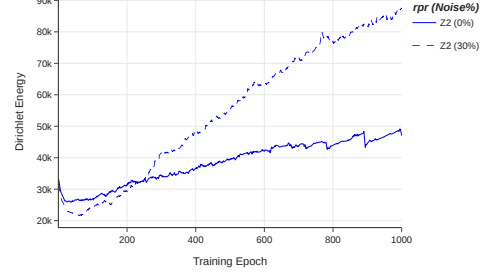
## 715 F Additional Empirical Studies on Energy Behavior

716 **Energy under clean overfitting vs. noise.** On ENZYMES without regularization, a deliberately  
 717 overfit GIN exhibits rising training accuracy and rising  $E_{\text{set}}^{\text{dir}}$ , confirming that energy tracks overfitting  
 718 in general. However, the rise under noise is markedly steeper.

719 **Energy scales with noise rate.** On the controlled PPA subset, we sweep symmetric noise  $\in$   
 720  $\{10, 20, 30, 40, 60\}\%$ . All curves follow the same U-shape (initial decrease, then rise), and the  
 721 asymptotic energy increases monotonically with noise rate. This monotonicity is what makes  $E_{\text{set}}^{\text{dir}}$  a  
 722 diagnostic.

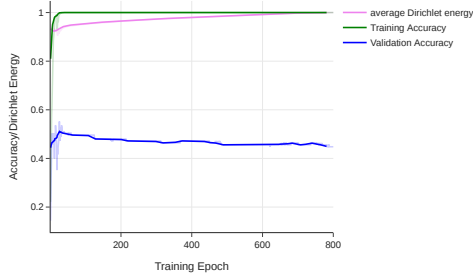


(a) Normalized  $E^{\text{dir}}(Y, Z_1, Z_2)$ .

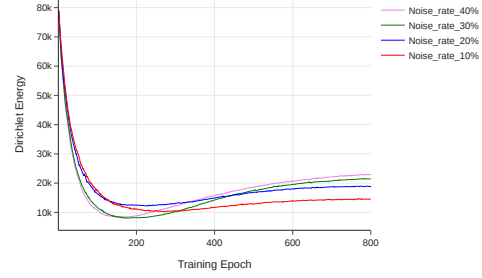


(b)  $E^{\text{dir}}(Z_2)$  alone.

Figure 4: HLFF-GNN frequency decomposition on ENZYMES. Solid lines: clean labels. Dashed: 30% symmetric noise. The high-frequency component  $Z_2$  is the only one whose energy diverges under noise, supporting the theoretical prediction.



(a) Clean overfitting on ENZYMES.



(b)  $E_{\text{set}}^{\text{dir}}$  vs. noise rate on PPA.

Figure 5: (a) Even without label noise, energy rises during overfitting; the surge under noise is much steeper. (b) Higher noise rates yield monotonically higher final energy.

## 723 G Additional Tables and Figures

Table 5: Sensitivity of GCOD to the learning rate  $\eta_u$  of the outlier parameter  $\mathbf{u}$ , under 40% asymmetric noise.

| Dataset  | $\eta_u = 1$ | $\eta_u = 0.1$ | % Change |
|----------|--------------|----------------|----------|
| PROTEINS | 76.19        | 75.89          | −0.39%   |
| MNIST    | 72.61        | 72.48          | −0.18%   |
| ENZYMES  | 69.81        | 68.33          | −2.12%   |
| IMDB-B   | 72.89        | 71.94          | −1.30%   |
| MUTAG    | 93.19        | 92.61          | −0.62%   |
| REDDIT   | 47.94        | 48.08          | +0.29%   |
| MSRC-21  | 95.57        | 95.45          | −0.13%   |

## 724 H Full ogbg-ppa: Do-No-Harm Check

725 The standard ogbg-ppa benchmark has 37 classes and  $\sim 158,000$  graphs of mean order  $\approx 240$ , so a  
 726 5-layer GIN with 300 hidden units is structurally *under-parameterized* for the task. We trained CE,  
 727  $\mathcal{L}_{DE}$  (Fixed), CE+W2 and GCOD on the full benchmark with 0%, 20% and 40% symmetric noise.  
 728 Two qualitative observations are robust across runs:

- 729 1. *Training accuracy on the noisy-labeled subset stays close to chance for all methods, including*  
 730 *standard CE.* The under-parameterized model never reaches a regime where it can memorize the  
 731 corrupted graph-level labels, consistent with our failure-mode analysis of Section 4: F1 (number  
 732 of classes), F2 (graph order) and F3 (training-set size) all push the full PPA out of the noise-fitting  
 733 regime.

Table 6: Performance of CE vs. GCOD with 20% symmetric label noise. Last column reports the absolute gain (GCOD–CE) on test accuracy.

| Dataset  | Metric | 0% CE | 20% CE | 20% GCOD | 20% (GCOD–CE) |
|----------|--------|-------|--------|----------|---------------|
| MNIST    | Best   | 72.69 | 66.30  | 71.26    | +4.96         |
|          | Last   | 69.80 | 53.64  | 67.88    | +14.24        |
|          | Diff   | 2.89  | 12.66  | 3.38     |               |
| ENZYMES  | Best   | 73.33 | 64.16  | 68.69    | +4.53         |
|          | Last   | 65.78 | 57.50  | 62.54    | +5.04         |
|          | Diff   | 7.55  | 6.66   | 6.15     |               |
| MSRC-21  | Best   | 96.69 | 90.26  | 94.69    | +4.43         |
|          | Last   | 93.80 | 79.64  | 90.26    | +10.62        |
|          | Diff   | 2.89  | 10.62  | 4.43     |               |
| PROTEINS | Best   | 81.16 | 76.23  | 79.38    | +3.15         |
|          | Last   | 79.18 | 62.32  | 78.12    | +15.80        |
|          | Diff   | 1.98  | 13.91  | 1.26     |               |
| MUTAG    | Best   | 94.73 | 89.47  | 90.01    | +0.54         |
|          | Last   | 84.21 | 68.42  | 86.84    | +18.42        |
|          | Diff   | 10.52 | 21.05  | 3.17     |               |
| IMDB-B   | Best   | 76.50 | 75.00  | 75.40    | +0.40         |
|          | Last   | 71.60 | 71.00  | 73.50    | +2.50         |
|          | Diff   | 4.90  | 4.00   | 1.90     |               |
| REDDIT   | Best   | 48.15 | 45.05  | 44.98    | −0.07         |
|          | Last   | 46.01 | 44.67  | 44.89    | +0.22         |
|          | Diff   | 2.14  | 0.38   | 0.09     |               |

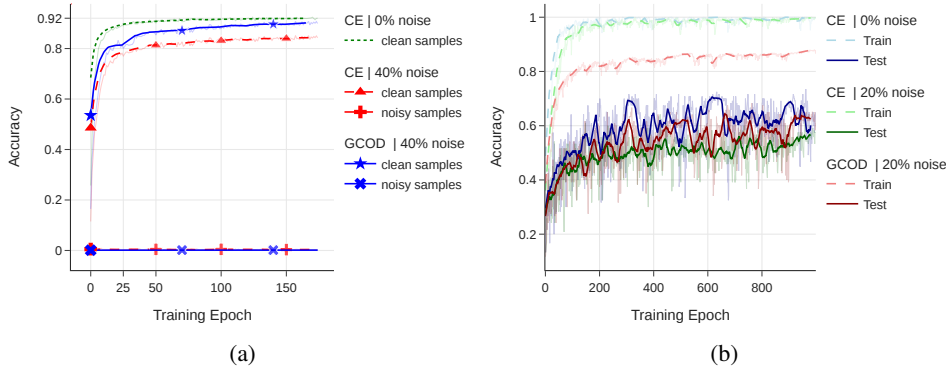


Figure 6: (a) Train accuracy on known noisy and clean PPA subsets (4-class, 40% symmetric noise) for GCN with CE: the model never separates noisy from clean training samples without a robust loss. (b) Train/test accuracy on ENZYMES with GIN under clean and 20% symmetric noise.

2. *Test accuracy of our three methods is statistically indistinguishable from CE at every noise rate.* None of CE+W2,  $\mathcal{L}_{DE}$ , or GCOD perturbs the underlying training dynamics enough to harm clean generalization, since their corrective signals (negative-eigenvalue projection, energy upper bound, outlier discounting) are only triggered by the high-frequency surge that does not occur on the full PPA.

This is by design: the methods are conditional regularizers, not unconditional smoothers. The take-away is that they impose *no measurable overhead in accuracy* when the underlying network is already noise-robust, while providing substantial protection when over-parameterization, low graph order, or small training sets push the network into the noise-fitting regime (Section 8, controlled PPA-30%-6 subset). A fair generality claim therefore rests on the seven additional benchmarks in Tables 2 and 6, with the full PPA serving exclusively as a do-no-harm sanity check.

## I Cross-Setting Transfer: Cora Node Classification

To assess whether the energy perspective captures a property of GNN training under label noise that is more general than the graph-classification setting it was designed for, we apply CE+W2 and GCOD

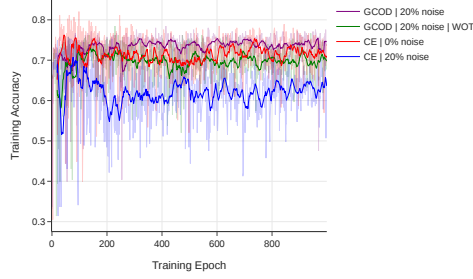


Figure 7: Ablation on the training-accuracy feedback in GCOD. Without scaling  $u$  by  $a_{\text{train}}$  (“GCOD WOT”), the outlier parameter dominates early and harms generalization; the full GCOD activates outlier-discounting gradually.

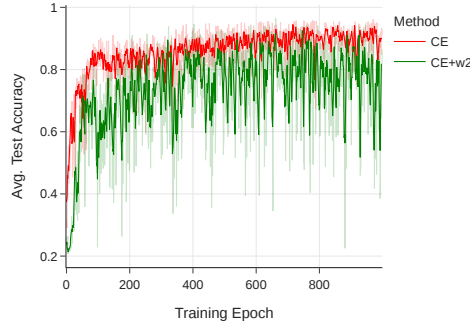


Figure 8: Test accuracy on PPA-30%-6 for CE and CE+W2. CE+W2 reaches comparable peak accuracy but exhibits unstable convergence due to the per-epoch eigendecomposition.

748 *unchanged* to node classification on Cora. We use both GAT and GCN backbones with the standard  
 749 public split, the transductive setting, and compare against 13 baselines: standard cross-entropy (CE;  
 750 column “standard”), UnionNet, GraphCleaner, GNNGuard, GNN-Cleaner, PI-GNN, NRGNN [Dai  
 751 et al., 2021], RTGNN [Yuan et al., 2023a], Community-Defense, ERASE, and CR-GNN. We evaluate  
 752 four label-noise regimes: clean (no noise), symmetric uniform 20%, symmetric random 20%, and  
 753 pairwise flip 20%. Each entry is mean $\pm$ std over 3 independent runs.

Table 7: Cora node classification, GAT backbone. Each cell is test accuracy (mean over 3 runs; std deviations available in the released code repository). Best per column is bold; our methods (CE+W2 and GCOD) are highlighted in gray. “–” denotes settings not run for the given baseline.

| Method                | Clean         | Flip 20%      | Random 20%    | Uniform 20%   |
|-----------------------|---------------|---------------|---------------|---------------|
| UnionNet              | <b>0.8287</b> | <b>0.7997</b> | <b>0.8183</b> | <b>0.8060</b> |
| CE+W2 (positive_eig.) | 0.8157        | 0.7660        | 0.7603        | 0.7483        |
| GraphCleaner          | 0.8023        | 0.7920        | 0.7877        | 0.7690        |
| GCOD (gcod)           | 0.8120        | 0.7607        | 0.7733        | 0.7870        |
| GNNGuard              | 0.7983        | 0.7527        | 0.7787        | 0.7460        |
| Community-Defense     | 0.7953        | 0.7440        | 0.7680        | 0.7390        |
| CE (standard)         | 0.7950        | 0.7440        | 0.7693        | 0.7437        |
| PI-GNN                | 0.7927        | 0.7303        | 0.7663        | 0.7390        |
| RTGNN                 | 0.7807        | 0.7247        | 0.7553        | 0.7203        |
| GNN-Cleaner           | 0.7790        | 0.6917        | –             | –             |
| ERASE                 | 0.7710        | 0.7350        | 0.7703        | 0.7483        |
| CR-GNN                | 0.7570        | 0.6803        | –             | –             |
| NRGNN                 | 0.6887        | 0.6207        | –             | –             |

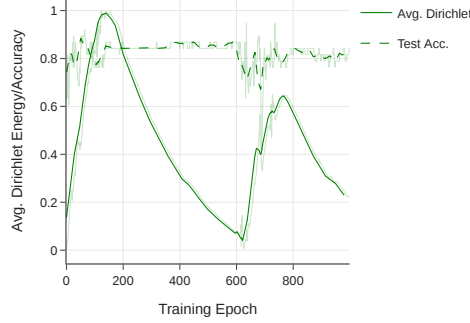


Figure 9: Average test accuracy and average within-graph  $E_{\text{set}}^{\text{dir}}$  on MUTAG (0% noise) with GIN: classification accuracy improves while energy decays, the clean-data signature of a healthy GNN.

Table 8: Cora node classification, GCN backbone. Each cell is test accuracy (mean over 3 runs). Best per column is bold; our methods are highlighted in gray.

| Method                | Clean         | Flip 20 %     | Random 20 %   |
|-----------------------|---------------|---------------|---------------|
| UnionNet              | <b>0.8060</b> | <b>0.7960</b> | <b>0.8040</b> |
| NRGNN                 | 0.7937        | 0.6790        | 0.7270        |
| GCOD (gcod)           | 0.7920        | 0.7687        | 0.7613        |
| Community-Defense     | 0.7903        | 0.7213        | 0.7573        |
| CE+W2 (positive_eig.) | 0.7880        | 0.7543        | 0.7743        |
| GraphCleaner          | 0.7860        | 0.7797        | 0.7700        |
| GNN-Cleaner           | 0.7760        | 0.6650        | 0.7050        |
| GNNGuard              | 0.7700        | 0.7073        | 0.7130        |
| CE (standard)         | 0.7700        | 0.7073        | 0.7130        |
| PI-GNN                | 0.7620        | 0.6830        | 0.7087        |
| RTGNN                 | 0.7543        | 0.7033        | 0.7017        |
| CR-GNN                | 0.7320        | 0.6100        | 0.6513        |
| ERASE                 | 0.6957        | 0.6597        | 0.6897        |

**Findings.** Tables 7 and 8 show that on every backbone-noise combination, at least one of CE+W2 or GCOD ranks in the top three among 13 methods, despite neither using neighbor labels, homophily, contrastive losses, pseudo-labels, nor any other node-specific machinery. Highlights: GCOD reaches 0.787 under uniform-20% on GAT (#2 of 10), 0.792 on clean GCN (#3 of 13), and 0.769 on flip-20% GCN (#3 of 13); CE+W2 reaches 0.816 on clean GAT (#2 of 13) and 0.774 on random-20% GCN (#3 of 13). The only baseline that consistently outperforms both is UnionNet, a node-classification-specific method that exploits label-propagation priors unavailable in the graph-classification setting our methods were designed for.

**Interpretation, consistent with Theorem 1.** The spectral mechanism by which noise fitting forces within-graph energy to grow is not specific to graph classification: in node classification, individual labels are also embedded in node-level representations whose Dirichlet energy reflects local smoothness, and constraining that smoothness penalizes the same noise-fitting path. The transfer is therefore not a coincidence but a corollary of the energy-based framing. Additional comparisons on db1p (GCN clean) and amazon-computers (GCN clean), available in the released code repository, show CE+W2 reaching 0.866 on db1p (#3 of 13) and competitive accuracy on amazon-computers, further supporting the cross-setting generality.



## 770 NeurIPS Paper Checklist

### 771 1. Claims

772 Question: Do the main claims made in the abstract and introduction accurately reflect the  
773 paper’s contributions and scope?

774 Answer: [\[Yes\]](#)

775 Justification: The abstract and Section 1 state precisely the claims supported by the theoreti-  
776 cal results in Section 6 and the experiments in Section 8.

### 777 2. Limitations

778 Question: Does the paper discuss the limitations of the work performed by the authors?

779 Answer: [\[Yes\]](#)

780 Justification: A dedicated paragraph in the Conclusion discusses the absence of a closed-  
781 form noise-energy characterization, reliance on synthetic noise, and the lack of standardized  
782 real-world noisy graph-classification benchmarks.

### 783 3. Theory assumptions and proofs

784 Question: For each theoretical result, does the paper provide the full set of assumptions and  
785 a complete (and correct) proof?

786 Answer: [\[Yes\]](#)

787 Justification: Proposition 1, Lemmas 1–2, Theorem 1, Lemma 3 and Corollary 1 state their  
788 assumptions explicitly in the main text, and full proofs are provided in Appendix B.

### 789 4. Experimental result reproducibility

790 Question: Does the paper fully disclose all the information needed to reproduce the main  
791 experimental results of the paper?

792 Answer: [\[Yes\]](#)

793 Justification: Appendix C provides datasets, splits, hyperparameters, architectures, and  
794 the implementation details of all three methods. Anonymized code is provided in the  
795 supplemental material.

### 796 5. Open access to data and code

797 Question: Does the paper provide open access to the data and code, with sufficient instruc-  
798 tions to faithfully reproduce the main experimental results?

799 Answer: [\[Yes\]](#)

800 Justification: All datasets are public benchmarks; an anonymous code repository accompa-  
801 nies the submission.

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803 Question: Does the paper specify all the training and test details necessary to understand the  
804 results?

805 Answer: [\[Yes\]](#)

806 Justification: Section 8 and Appendix C report all training and evaluation details.

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809 information about the statistical significance of the experiments?

810 Answer: [\[Yes\]](#)

811 Justification: All main results report mean $\pm$ std over 4 independent runs with different seeds.

### 812 8. Experiments compute resources

813 Question: For each experiment, does the paper provide sufficient information on the com-  
814 puter resources?

815 Answer: [\[Yes\]](#)

816 Justification: Appendix C reports the use of NVIDIA RTX A6000 GPUs and Table 4 reports  
817 relative training-time overheads.

818 **9. Code of ethics**

819 Question: Does the research conducted in the paper conform, in every respect, with the

820 NeurIPS Code of Ethics?

821 Answer: [Yes]

822 Justification: The work uses public benchmarks, does not involve human subjects, and the

823 broader-impact paragraph discusses application contexts.

824 **10. Broader impacts**

825 Question: Does the paper discuss both potential positive societal impacts and negative

826 societal impacts of the work performed?

827 Answer: [Yes]

828 Justification: Discussed in the Conclusion. The work primarily benefits scientific applica-

829 tions (chemistry, biology) where label noise is systematic.

830 **11. Safeguards**

831 Question: Does the paper describe safeguards that have been put in place for responsible

832 release of data or models that have a high risk for misuse?

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834 Justification: The paper does not release new datasets or pretrained models with misuse risk.

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837 and terms of use explicitly mentioned and properly respected?

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841 **13. New assets**

842 Question: Are new assets introduced in the paper well documented and is the documentation

843 provided alongside the assets?

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847 Question: For crowdsourcing experiments and research with human subjects, does the paper

848 include the full text of instructions given to participants and screenshots, if applicable, as

849 well as details about compensation (if any)?

850 Answer: [N/A]

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852 **15. Institutional review board (IRB) approvals or equivalent for research with human**

853 **subjects**

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855 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

856 approvals (or an equivalent approval/review based on the requirements of your country or

857 institution) were obtained?

858 Answer: [N/A]

859 Justification: No human subjects.

860 **16. Declaration of LLM usage**

861 Question: Does the paper describe the usage of LLMs if it is an important, original, or

862 non-standard component of the core methods in this research?

863 Answer: [N/A]

864 Justification: LLMs are not part of the core methods.